Innovative Product Development

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Title: Learnova — AI-powered redesign of outdated lecture slides into interactive, illustrated teaching decks

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IDEA/APPROACH DETAILS

❖ Problem Statement

Lecture slides used in colleges are mostly old PowerPoint files or PDFs: dense bullet text, equations and diagrams pasted as pictures, missing titles and no interaction. Faculty do not have the time to redesign every deck, and AI deck generators create new slides from a prompt but rewrite or drop the teacher's content and cannot redesign an existing deck faithfully.

Learnova redesigns an existing lecture deck into a modern, illustrated, interactive deck while keeping every point the teacher wrote.

❖ Objectives

1. Read real lecture material (PPTX, slide PDFs, Word notes, typed topics) and recover titles, bullet levels, tables, figures and picture-only slides.
2. Choose a suitable visual for each slide (process, timeline, comparison, definition, chart …) and draw an illustration on every slide.
3. Keep 100% of the source content: no item dropped, no fact invented, no word broken across lines.
4. Make slides interactive: checkpoint quizzes, step-by-step solved problems and hands-on labs for hard topics (NLP, ML, IPCV).
5. Work with AI keys for better titles, explanations and diagrams, and fully offline when keys are unavailable.

Literature Review/Existing Systems

Sr. No.	Reference Paper / Existing Application	Key Findings	Gap / Scope for Improvement
1	Hu & Wan, “PPSGen: Learning-based presentation slides generation for academic papers”, IEEE TKDE, 2015	Regression model scores sentence importance; ILP selects sentences into slides.	Works on papers only; summarises (loses content); no visuals, no redesign of existing slides.
2	Fu et al., “DOC2PPT: Automatic presentation slides generation from scientific documents”, AAAI, 2022	Neural summariser + figure selection + layout prediction, trained on paper–slide pairs.	Needs training data; abstractive text can drop or alter facts; input must be a paper.
3	Sun et al., “D2S: Document-to-slide generation via query-based text summarization”, NAACL, 2021	Slide titles used as queries; QA-style retrieval + summarisation per slide.	Text only; no figures or interactivity; cannot rebuild an existing deck.
4	Garner & Alley, “How the design of presentation slides affects audience comprehension: assertion–evidence approach”, IJEE, 2013	Sentence headlines + visual evidence improve comprehension over bullet lists.	Design guidance only — no tool applies it automatically to old decks.
5	Existing tools: Gamma, Microsoft Copilot in PowerPoint, Gemini in Slides, Canva Magic Design	LLM outline → template fill; good-looking new decks from a prompt.	Rewrite/merge the teacher's content, lose original figures, no quizzes/labs, need internet and paid AI.

Common limitations: existing systems generate new decks instead of redesigning old ones, are not faithful to the source, ignore picture-only and equation slides, and offer no interactivity or offline mode.

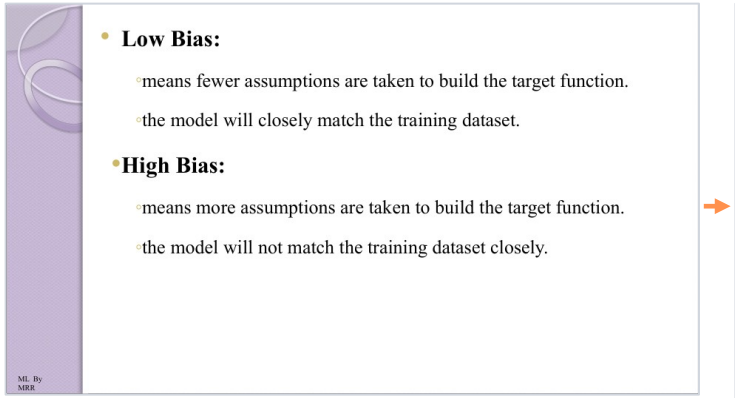
IDEA

Prototype — upload an old deck, get a redesigned interactive deck

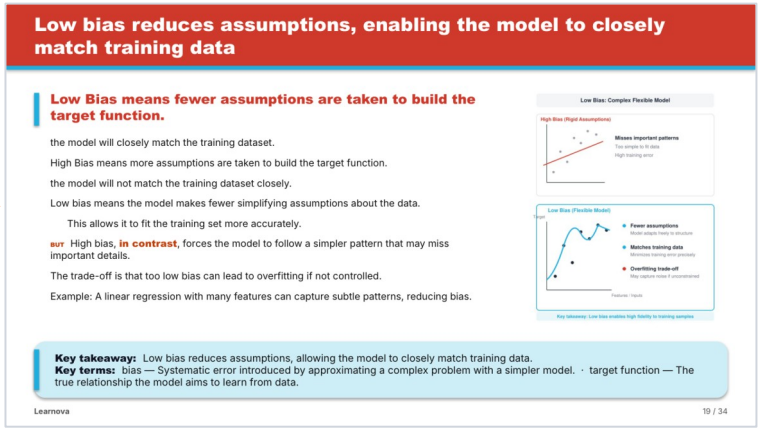
How it works

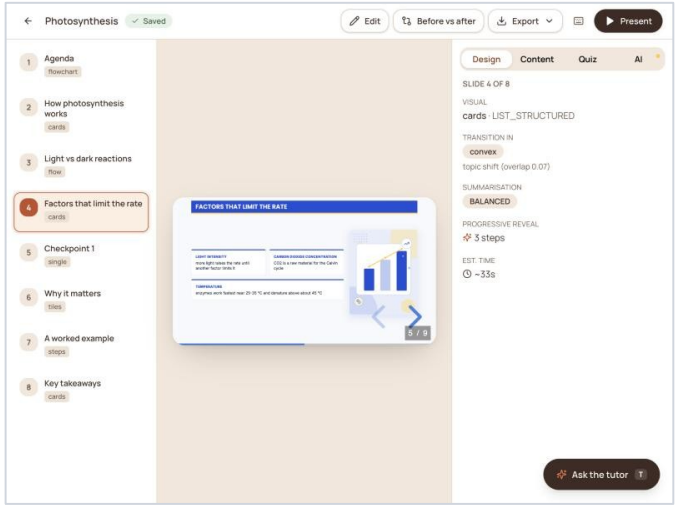
1. Upload a lecture deck — PPTX, PDF, Word notes — or type a topic.
2. Learnova reads every slide: titles, bullet levels, tables, figures; OCR for picture slides.
3. Each slide gets the right visual, an assertion title, and an illustration (AI-drawn diagram, the teacher's figure, or an offline illustration).
4. Quizzes, step-by-step solved problems and labs are added.
5. Download an editable PowerPoint, an interactive web deck and a study pack; share a link.

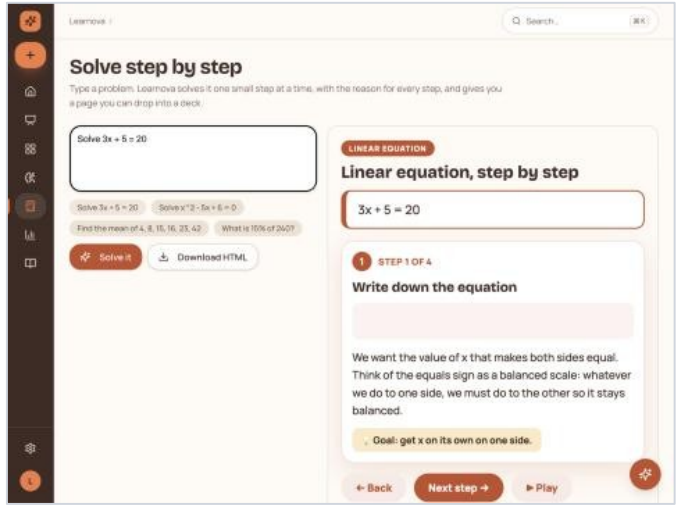
Before — original slide



After — Learnova slide

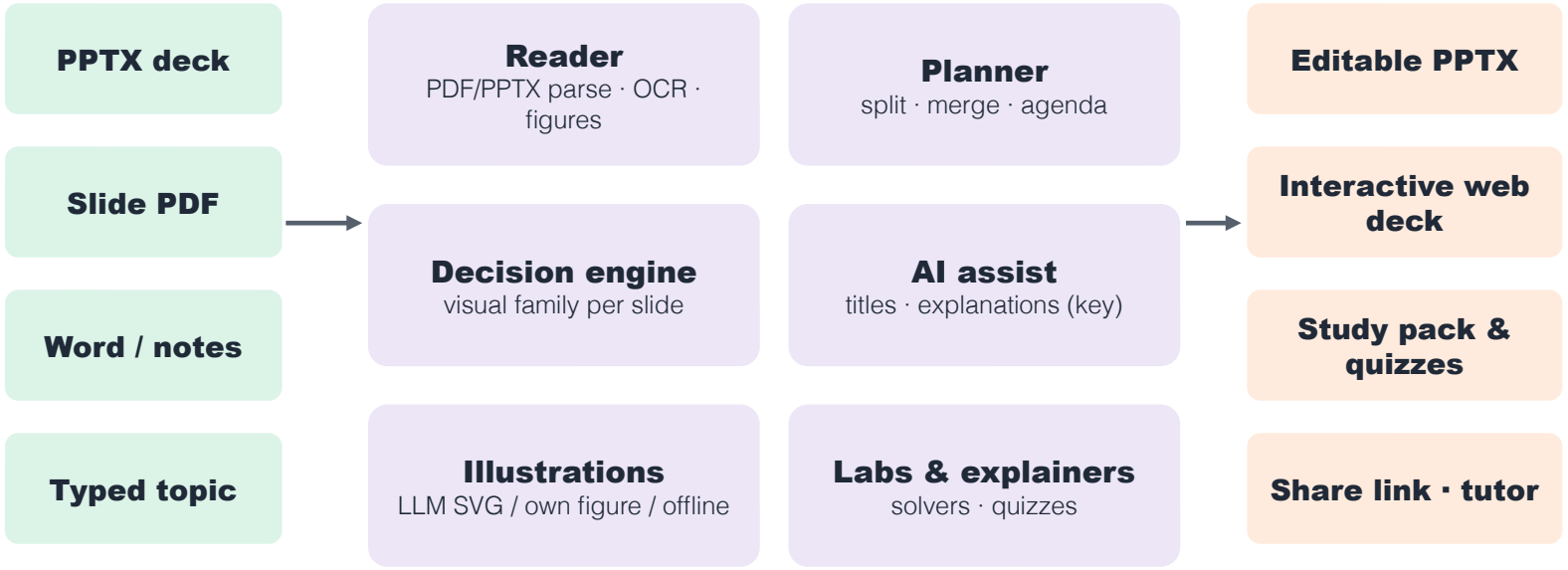






ARCHITECTURE

Input → Processing (workflow) → **Output**



Layout engine + audit: text measured glyph by glyph — no overflow, no broken words

FastAPI service · job queue · deck library · OTP sign-in · AI router (Gemini ×5 → Groq → NVIDIA → offline rules)
React front-end: Create · Generate (live progress) · Preview · Present · Solve · Export · Settings

TECHNOLOGY STACK

- **Front-end:** React 18, Vite, Tailwind CSS, Reveal.js (web deck), installable PWA
- **Back-end:** Python 3.12, FastAPI, background job queue, SQLite
- **Document processing:** python-pptx, PyMuPDF, LibreOffice (PPTX→PDF), Tesseract OCR, python-docx
- **AI / NLP:** Google Gemini (5-key pool), Groq, NVIDIA NIM — router with failover; rule-based offline engine
- **Maths & labs:** SymPy (exact solvers), vanilla JS lab engine, SVG illustrations
- **Quality:** pytest (750+ tests), layout audit, gold-set evaluation

USE CASES

Where Learnova is used

Faculty before a lecture

Uploads last year's PPT; gets a redesigned, illustrated deck with quizzes in under a minute, fully editable in PowerPoint.

Students revising for exams

Open the web deck, flip key-term cards, solve checkpoint questions and step through solved problems at their own pace.

Lab-heavy subjects

IPCV / NLP / ML topics get hands-on labs: region growing, split & merge, CYK, Viterbi, Naive Bayes WSD, HMM tagging.

Colleges without paid AI

Everything works offline with rule-based logic; AI keys only improve titles, explanations and diagrams.

Remote and flipped classes

Share a view-only link; the deck tutor answers questions from the deck's own content.

Accessibility

Large readable text, no broken words, plain-language explanations and alt text for every illustration.

Implementation

Modules developed and integrated

Module	What is implemented	Status
Input readers	PPTX via LibreOffice → slide reader; PDF page reader with OCR; DOCX; typed topics	Done
Content Decision Engine	27 detectors, 20+ visual families, reason for every choice	Done
PowerPoint + web builders	Native editable shapes; Reveal.js deck with click-by-click reveals	Done
Illustrations	AI SVG diagram per slide (5 Gemini keys) · source figure · offline scenes; HTML link	Done
Quizzes & explainers	Checkpoint MCQs; exact solvers incl. Bayes, bigram, Naive Bayes, Viterbi, CYK	Done
Interactive labs	7 built-in (incl. Region Growing, Split & Merge) + generated labs per topic	Done
Platform	FastAPI jobs, OTP login, library, share links, study pack, PWA front-end	Done
Evaluation & papers	Gold-set accuracy, layout audit, fact retention; IEEE + Springer papers	Done
Classroom study	Controlled study with students on learning outcomes	Planned

Novelty — what Learnova does that others do not

Capability	Learnova	Gamma	Copilot PPT	Gemini Slides	Canva
Redesigns an existing deck (not only new ones)	✔	✘	~	~	~
Keeps every source point (fact check)	✔	✘	✘	✘	✘
Keeps teacher's figures, OCR of picture slides	✔	✘	~	✘	✘
Illustration on every slide + HTML link	✔	~	~	~	~
Exact step-by-step solved problems	✔	✘	✘	✘	✘
Interactive labs for hard topics	✔	✘	✘	✘	✘
Works offline without AI keys	✔	✘	✘	✘	✘
Explains every design choice	✔	✘	✘	✘	✘

✔ supported ~ partly (e.g. generic stock images, or only with a paid plan / internet) ✘ not supported — based on the team's comparison of each tool's documented features (Oct 2026).

Results on real course material

97%
visual chosen correctly
101 gold slides, 3 held-out sets

98–100%
facts kept
6 real decks, 36–216 pages

0
broken words / overflow
layout audit, all decks

26 / 26
problems solved correctly
held-out explainer benchmark

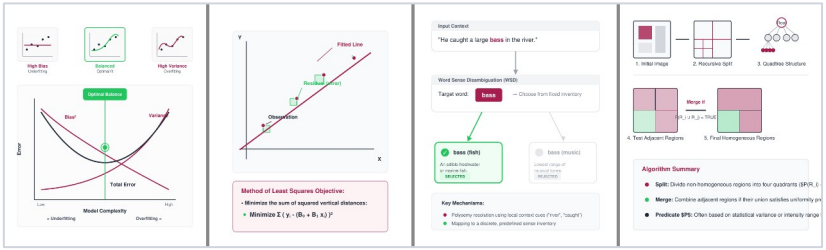
< 1 ms
decision per slide
rule-based engine

~35 s
216-page PDF → full deck
without any API key

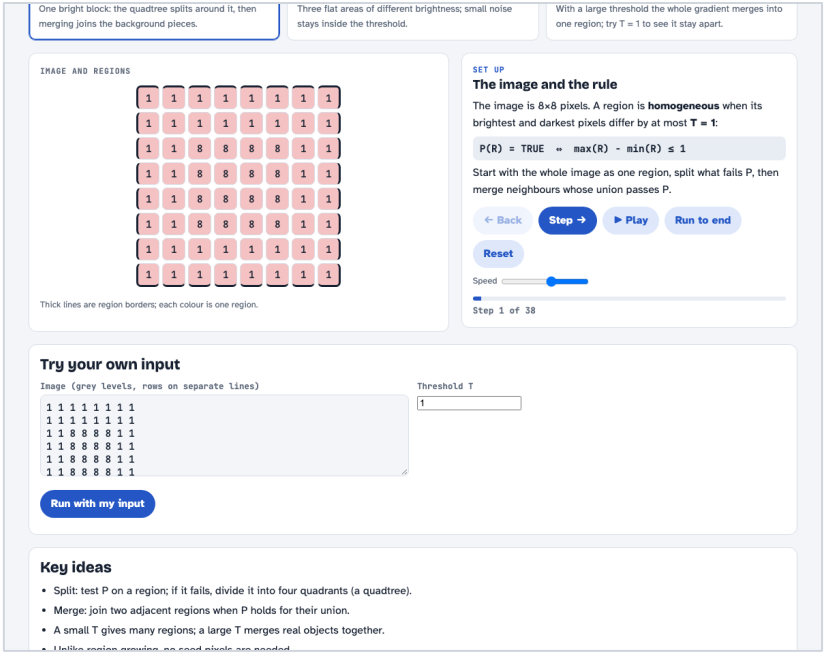
Tested on: NLP Module 4 (36 pp), Module 5 (95 pp), Syntax analysis (216 pp), WSD slides and notes, Lesk algorithm notes, ML Bias & Variance (PPTX, 28 slides) and Linear Regression (PPTX, 63 slides).

Linear Regression PPTX: earlier parser kept 23 of 63 slides; the new reader keeps all of them with the original derivations as figures.

AI-drawn diagrams (one per slide)



Region Splitting & Merging lab (IPCV)



Future scope and conclusion

Future scope

- Classroom study with students to measure learning gains.
- Narrated video export of each deck (text-to-speech).
- More labs: CNN filters, decision trees, gradient descent.
- Teacher review mode: approve or edit AI suggestions slide by slide.
- Hindi and regional-language decks.
- College LMS integration (Moodle / Google Classroom).

Conclusion

Learnova turns the decks teachers already have into clear, illustrated and interactive lessons without losing their content.

It is faithful by design (every point kept and checked), visual on every slide, interactive through quizzes, solved problems and labs, and dependable — it works with AI keys and without them.

Two research papers (IEEE conference format and Springer LNCS) describe the methods and the evaluation.

References:

Ref. Papers:

[1] Y. Hu and X. Wan, “PPSGen: Learning-based presentation slides generation for academic papers,” IEEE Trans. Knowl. Data Eng., vol. 27, no. 4, pp. 1085–1097, 2015.

[2] T.-J. Fu, W. Y. Wang, D. McDuff and Y. Song, “DOC2PPT: Automatic presentation slides generation from scientific documents,” in Proc. AAAI, vol. 36, pp. 634–642, 2022.

[3] E. Sun, Y. Hou, D. Wang, Y. Zhang and N. X. R. Wang, “D2S: Document-to-slide generation via query-based text summarization,” in Proc. NAACL, pp. 1405–1418, 2021.

[4] J. K. Garner and M. P. Alley, “How the design of presentation slides affects audience comprehension,” Int. J. Eng. Educ., vol. 29, no. 6, pp. 1564–1579, 2013.

[5] R. E. Mayer, Multimedia Learning, 2nd ed. Cambridge University Press, 2009.

[6] J. Sweller, “Cognitive load during problem solving: Effects on learning,” Cognitive Science, vol. 12, no. 2, pp. 257–285, 1988.

[7] J. Maynez, S. Narayan, B. Bohnet and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in Proc. ACL, pp. 1906–1919, 2020.

[8] R. Smith, “An overview of the Tesseract OCR engine,” in Proc. ICDAR, pp. 629–633, 2007.

[9] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed. draft, 2025.

References Web resources:

Gamma — https://gamma.app · Microsoft Copilot in PowerPoint · Google Gemini in Slides · Canva Magic Design — accessed Oct 2026

Reveal.js — https://revealjs.com · python-pptx — https://python-pptx.readthedocs.io · PyMuPDF — https://pymupdf.readthedocs.io

Project repository — https://github.com/SDP42/LearnovaV2